

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$3x = 36y - 45$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

Answer

- A. $x = 4y$
- B. $\frac{1}{3}x = 4y$
- C. $x = 12y - 15$
- D. $\frac{1}{3}x = 12y - 15$

Correct Answer: B

Rationale

Choice B is correct. A system of two linear equations in two variables, x and y , has no solution when the lines in the xy -plane representing the equations are parallel and distinct. Two lines are parallel and distinct if their slopes are the same and their y -intercepts are different. The slope of the graph of the given equation, $3x = 36y - 45$, in the xy -plane can be found by rewriting the equation in the form $y = mx + b$, where m is the slope of the graph and $(0, b)$ is the y -intercept. Adding 45 to each side of the given equation yields $3x + 45 = 36y$. Dividing each side of this equation by 36 yields $\frac{1}{12}x + \frac{5}{4} = y$, or $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the slope of the graph of the given equation is $\frac{1}{12}$ and the y -intercept is $(0, \frac{5}{4})$. Therefore, the graph of the second equation in the system must also have a slope of $\frac{1}{12}$, but must not have a y -intercept of $(0, \frac{5}{4})$. Multiplying each side of the equation given in choice B by $\frac{1}{4}$ yields $\frac{1}{12}x = y$, or $y = \frac{1}{12}x$. It follows that the graph representing the equation in choice B has a slope of $\frac{1}{12}$ and a y -intercept of $(0, 0)$. Since the slopes of the graphs of the two equations are equal and the y -intercepts of the graphs of the two equations are different, the equation in choice B could be the second equation in the system.

Choice A is incorrect. This equation can be rewritten as $y = \frac{1}{4}x$. It follows that the graph of this equation has a slope of $\frac{1}{4}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Choice C is incorrect. This equation can be rewritten as $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{12}$ and a y -intercept of $(0, \frac{5}{4})$, so the system consisting of this equation and the given equation has infinitely many solutions, rather than no solution.

Choice D is incorrect. This equation can be rewritten as $y = \frac{1}{36}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{36}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.